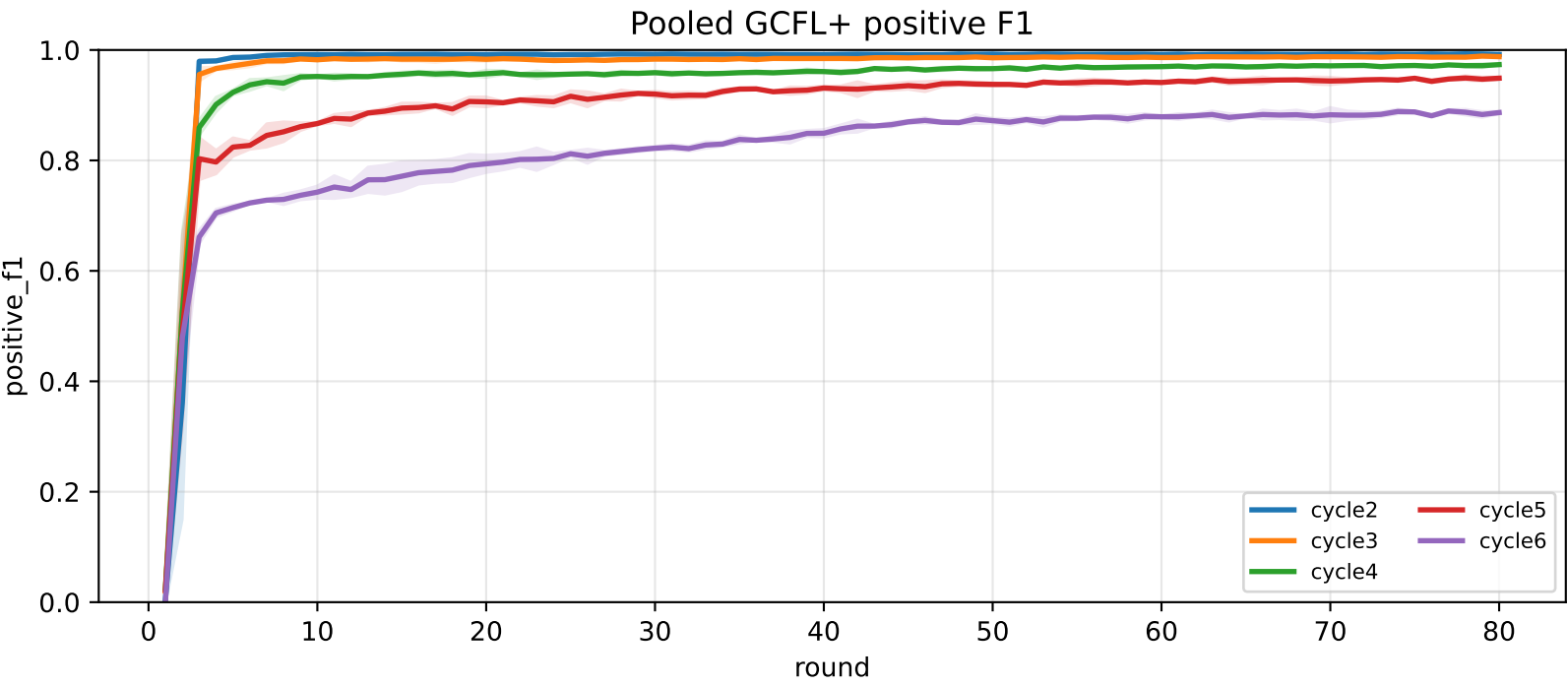
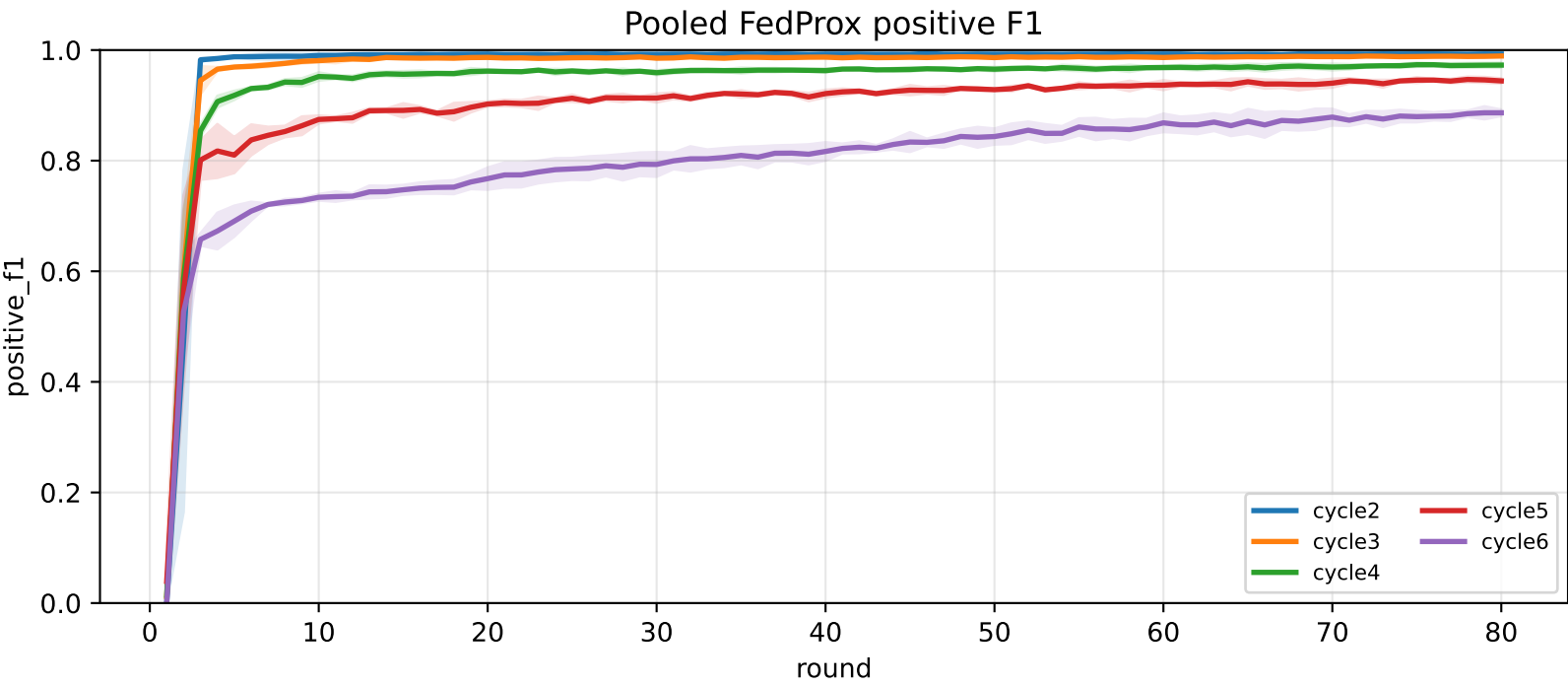
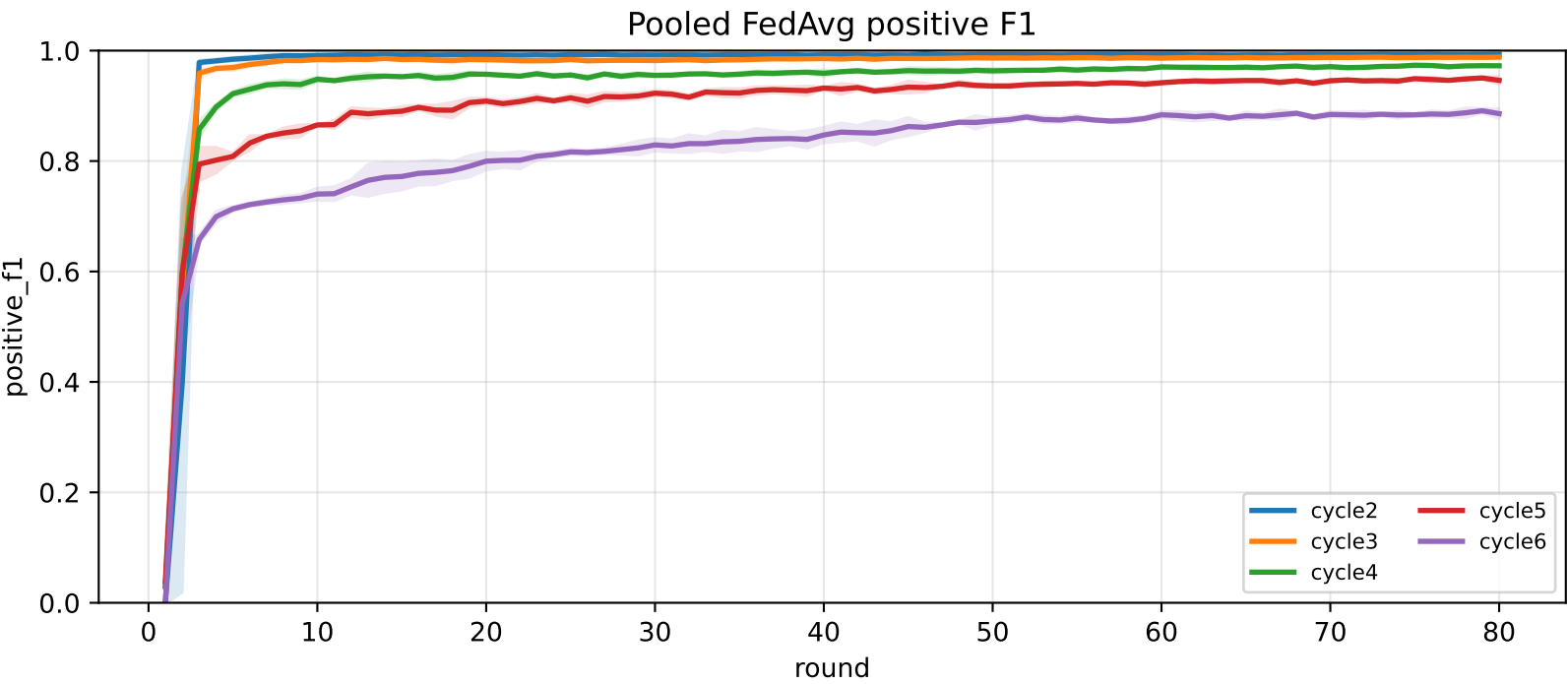
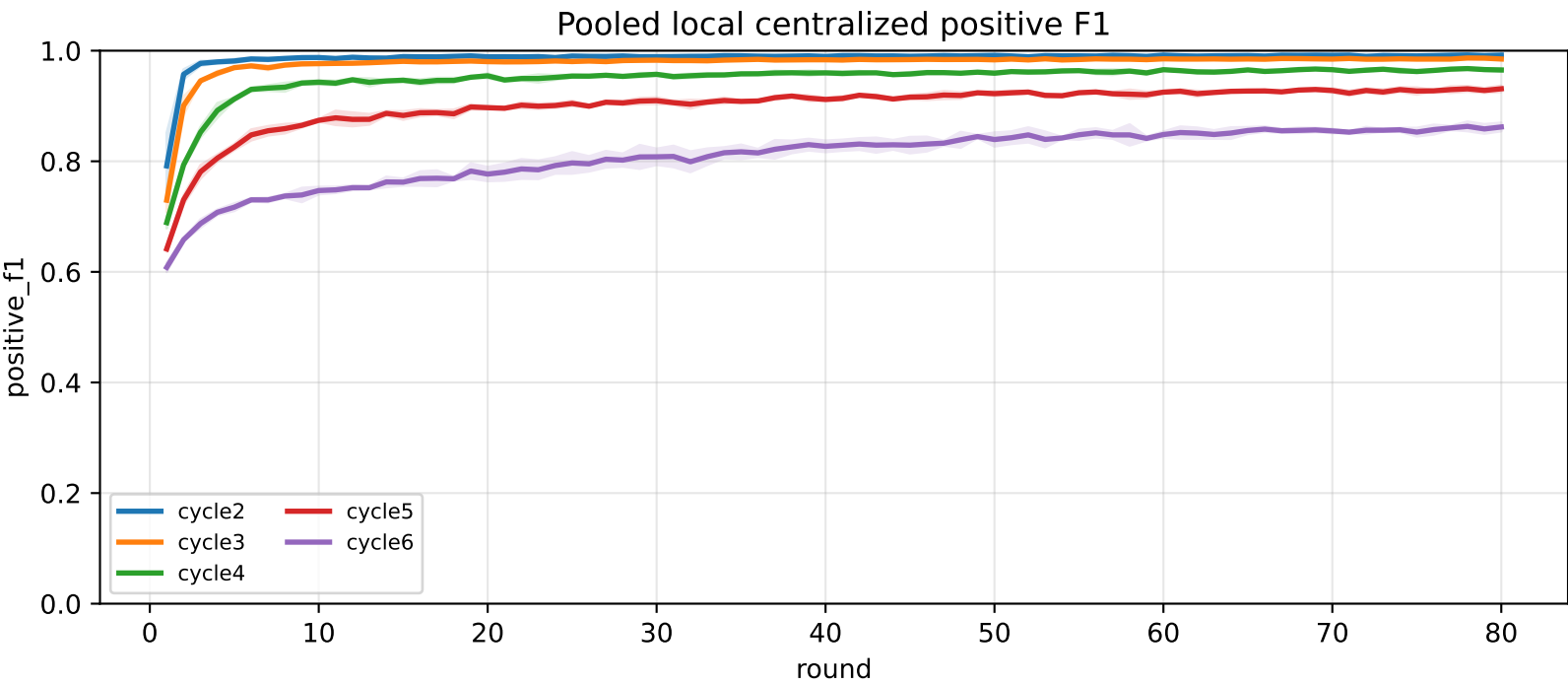
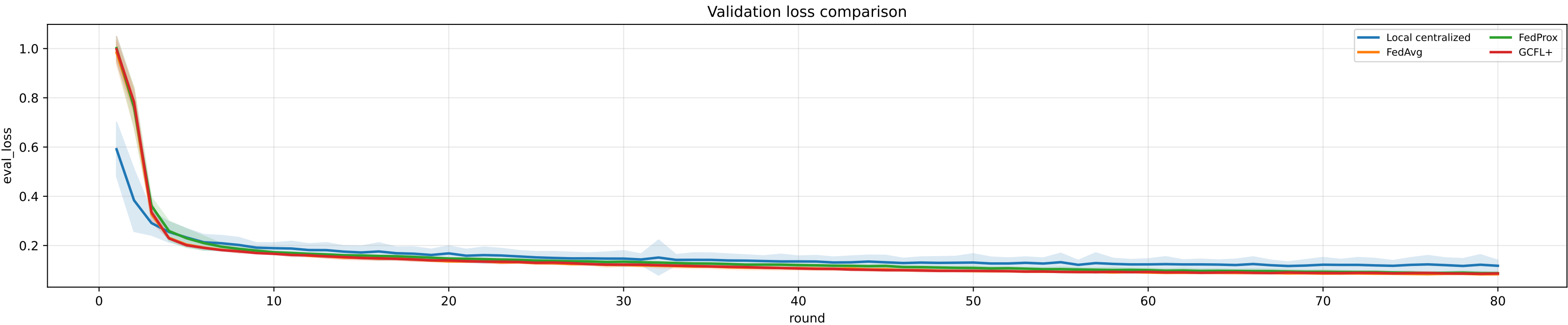


Heterogeneity level: Low | aggregated / pooled performance over 20 clients |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

heterogeneity=low | task_jsd_mean=0.009 | task_jsd_median=0.004 | task_jsd_max=0.075 | family_centroid_jsd_mean=0.009
per-task pos-rate std | cycle2_std=0.073 | cycle3_std=0.042 | cycle4_std=0.021 | cycle5_std=0.032 | cycle6_std=0.050

Best logged test performance (mean $\pm$ std over seeds)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	94.7% $\pm$ 0.2%	94.4% $\pm$ 0.2%	99.2% $\pm$ 0.0%	98.6% $\pm$ 0.1%	96.8% $\pm$ 0.2%	93.1% $\pm$ 0.2%	86.2% $\pm$ 0.5%
FedAvg	95.9% $\pm$ 0.1%	95.7% $\pm$ 0.1%	99.2% $\pm$ 0.1%	98.7% $\pm$ 0.1%	97.4% $\pm$ 0.1%	95.2% $\pm$ 0.2%	89.5% $\pm$ 0.2%
FedProx	95.8% $\pm$ 0.3%	95.5% $\pm$ 0.3%	99.2% $\pm$ 0.0%	98.8% $\pm$ 0.0%	97.4% $\pm$ 0.3%	94.7% $\pm$ 0.4%	89.2% $\pm$ 0.7%
GCFL+	95.6% $\pm$ 0.2%	95.3% $\pm$ 0.2%	99.3% $\pm$ 0.0%	98.7% $\pm$ 0.1%	97.2% $\pm$ 0.3%	94.8% $\pm$ 0.4%	88.4% $\pm$ 0.4%



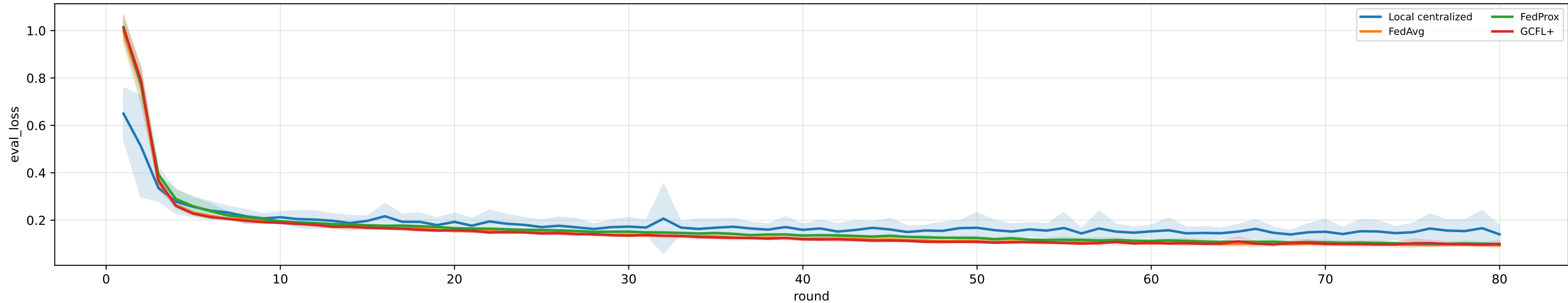
Heterogeneity level: Low | family aggregated / pooled performance |
family=strong_decreasing | n_clients=4 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=strong_decreasing | cycle2=35.9%±5.1% | cycle3=31.5%±1.3% | cycle4=25.3%±2.1% | cycle5=19.6%±3.4% | cycle6=15.3%±3.9%
heterogeneity=low | task_jsd_mean=0.009 | task_jsd_median=0.004 | task_jsd_max=0.075 | family_centroid_jsd_mean=0.009
per-task pos-rate std | cycle2_std=0.073 | cycle3_std=0.042 | cycle4_std=0.021 | cycle5_std=0.032 | cycle6_std=0.050

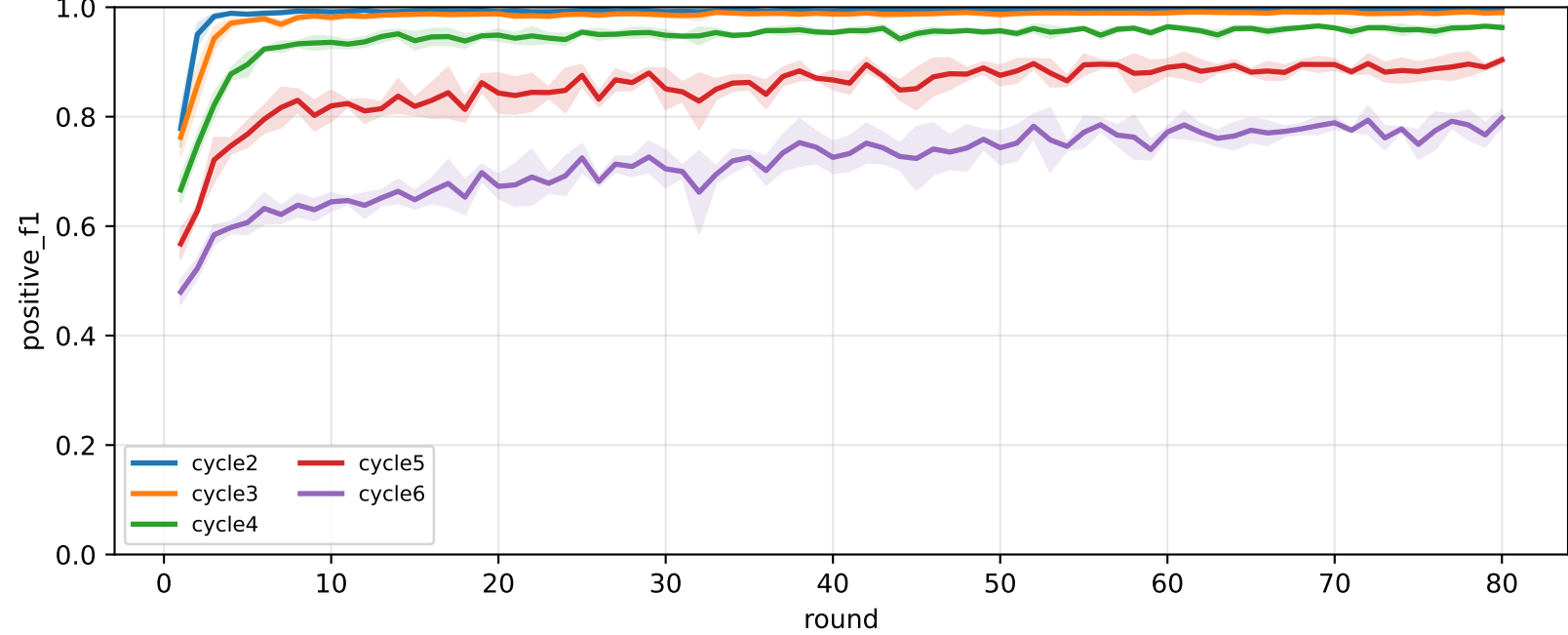
Best logged test performance (mean ± std over seeds)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	94.2% ± 0.3%	91.8% ± 0.3%	99.6% ± 0.1%	99.0% ± 0.1%	96.5% ± 0.2%	90.1% ± 1.0%	76.7% ± 0.6%
FedAvg	95.4% ± 0.0%	93.2% ± 0.0%	99.6% ± 0.1%	99.2% ± 0.1%	97.1% ± 0.2%	92.7% ± 0.2%	80.3% ± 0.3%
FedProx	95.3% ± 0.5%	93.2% ± 0.7%	99.7% ± 0.1%	99.3% ± 0.1%	97.5% ± 0.4%	92.5% ± 1.1%	79.8% ± 1.7%
GCFL+	95.0% ± 0.2%	92.8% ± 0.2%	99.6% ± 0.0%	99.3% ± 0.1%	97.2% ± 0.4%	92.1% ± 0.7%	78.8% ± 0.7%

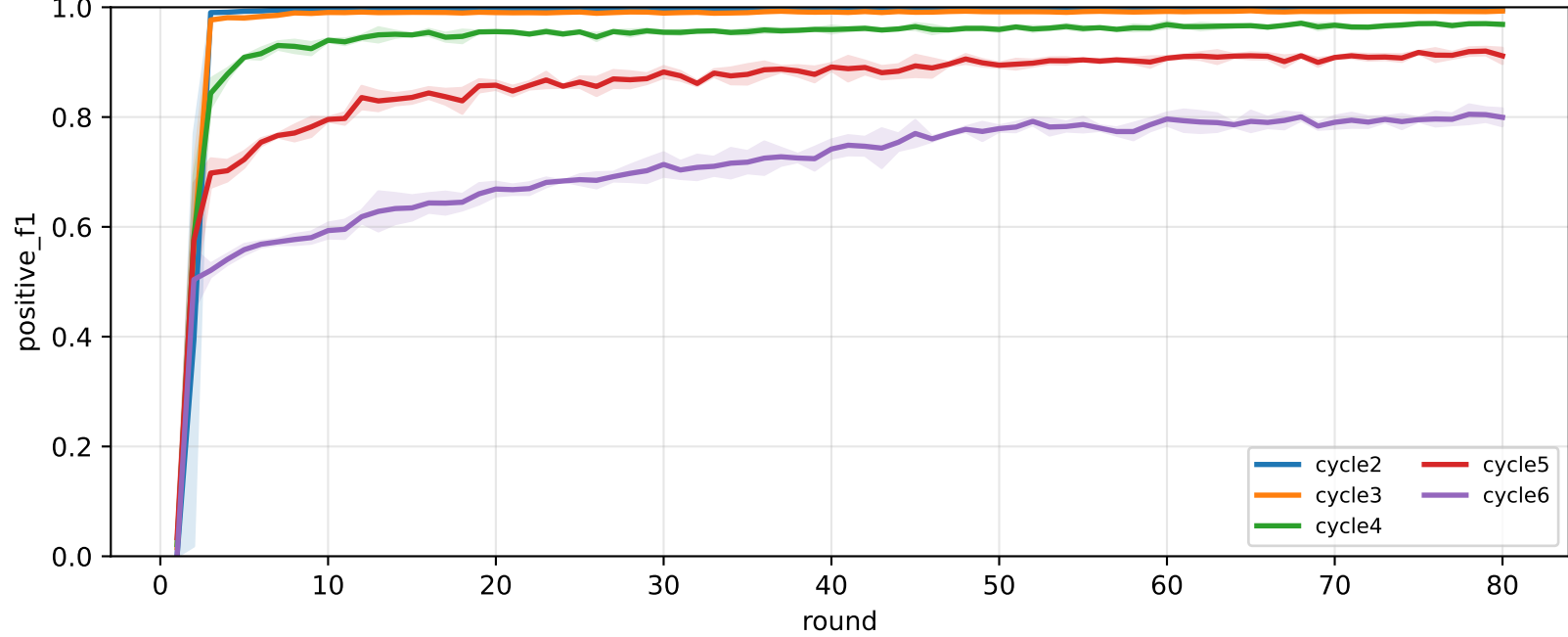
Validation loss comparison



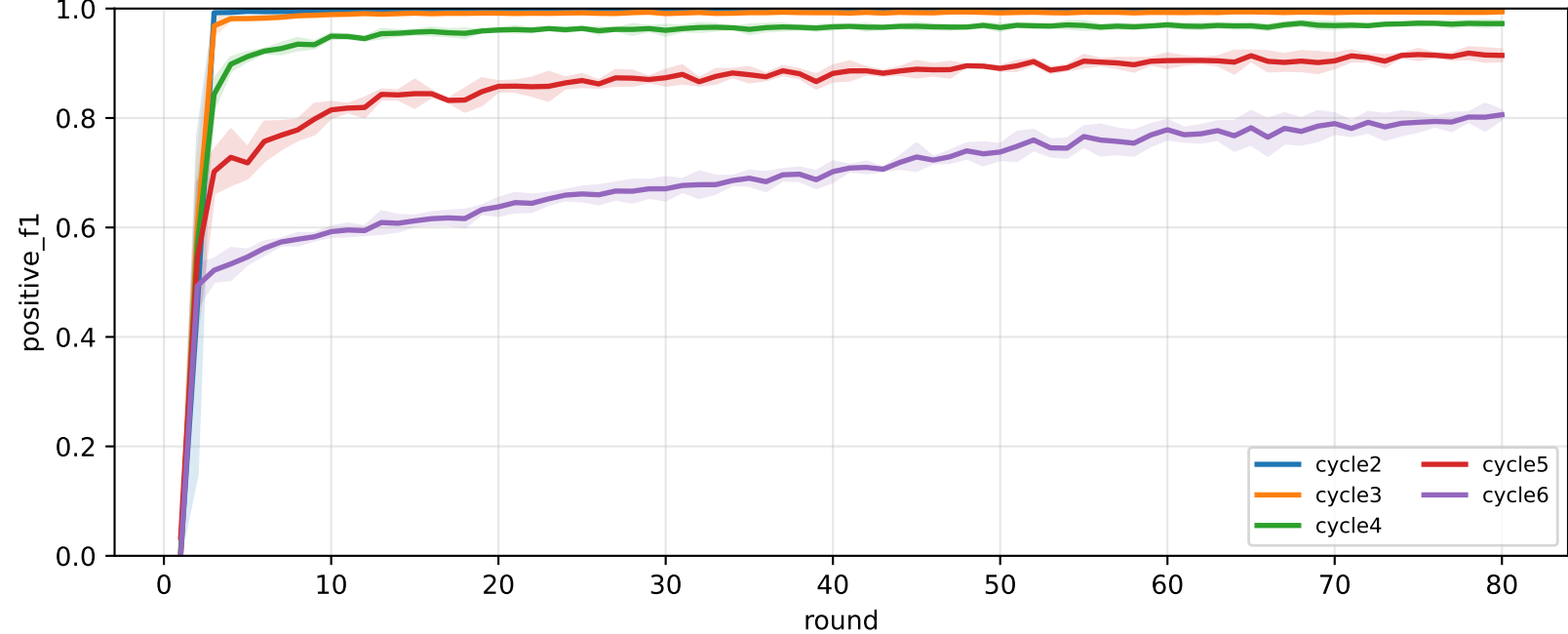
Pooled local centralized positive F1



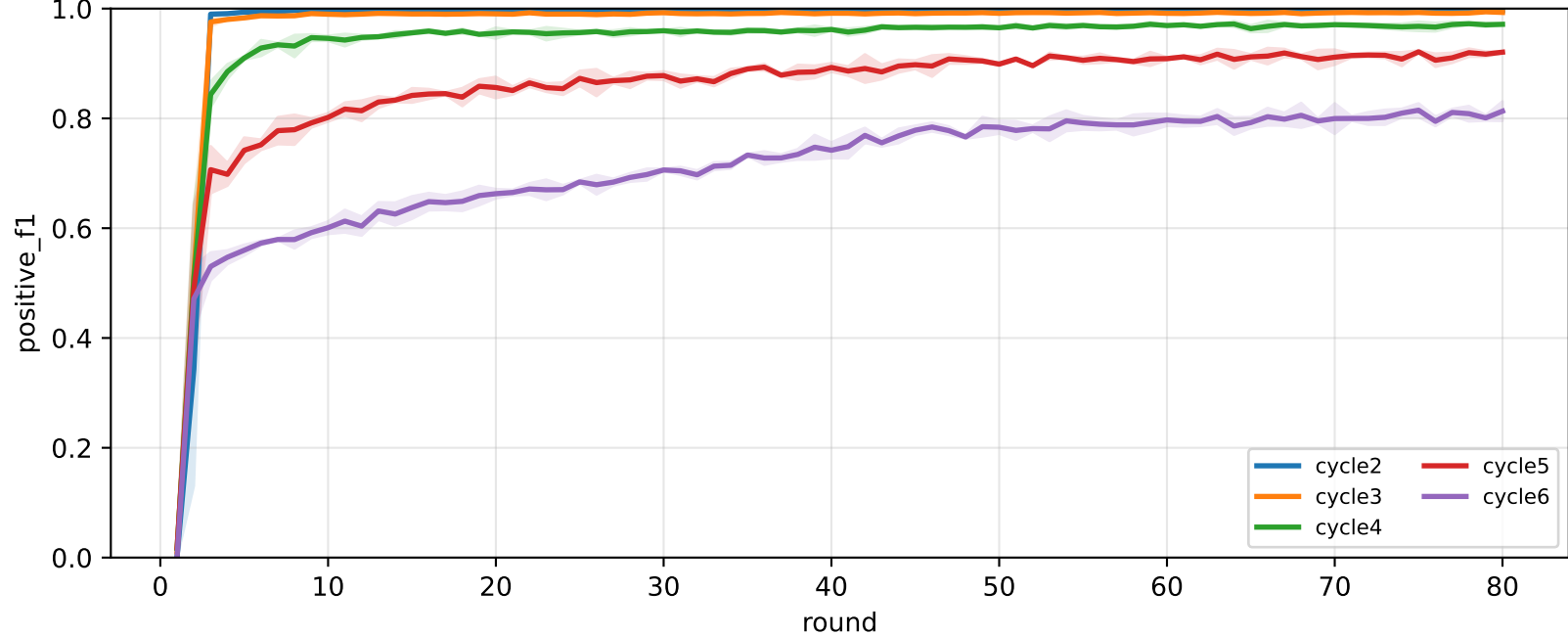
Pooled FedAvg positive F1



Pooled FedProx positive F1



Pooled GCFL+ positive F1



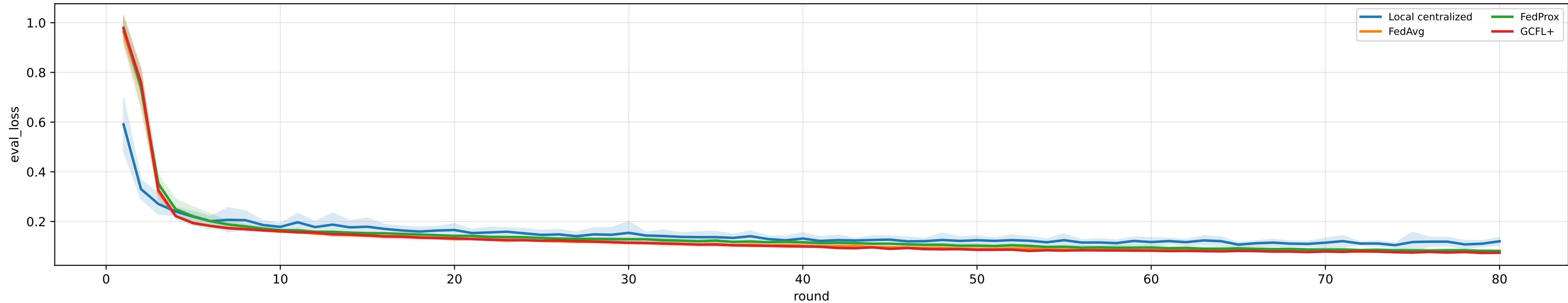
Heterogeneity level: Low | family aggregated / pooled performance |
family=mild_decreasing | n_clients=4 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=mild_decreasing | cycle2=28.4%±0.4% | cycle3=28.5%±0.9% | cycle4=28.1%±0.7% | cycle5=25.6%±0.3% | cycle6=23.4%±0.3%
heterogeneity=low | task_jsd_mean=0.009 | task_jsd_median=0.004 | task_jsd_max=0.075 | family_centroid_jsd_mean=0.009
per-task pos-rate std | cycle2_std=0.073 | cycle3_std=0.042 | cycle4_std=0.021 | cycle5_std=0.032 | cycle6_std=0.050

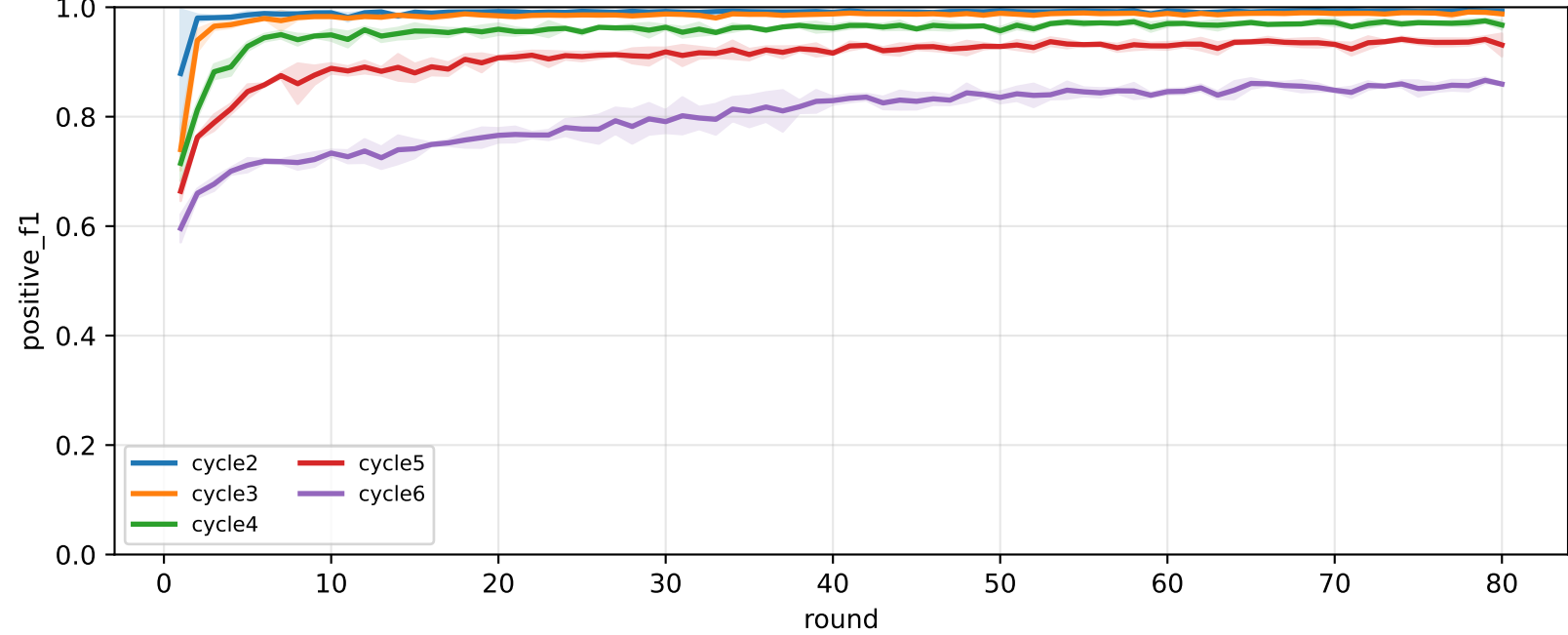
Best logged test performance (mean ± std over seeds)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	95.3% ± 0.2%	95.2% ± 0.2%	99.3% ± 0.1%	98.9% ± 0.1%	97.4% ± 0.3%	93.3% ± 0.4%	86.9% ± 0.7%
FedAvg	96.4% ± 0.2%	96.2% ± 0.2%	99.3% ± 0.0%	99.0% ± 0.1%	97.9% ± 0.3%	95.6% ± 0.3%	89.4% ± 0.2%
FedProx	96.3% ± 0.2%	96.1% ± 0.2%	99.3% ± 0.0%	99.1% ± 0.1%	98.0% ± 0.1%	95.1% ± 0.2%	89.0% ± 0.7%
GCFL+	96.2% ± 0.0%	96.1% ± 0.0%	99.4% ± 0.0%	99.1% ± 0.1%	98.0% ± 0.1%	95.5% ± 0.1%	88.4% ± 0.2%

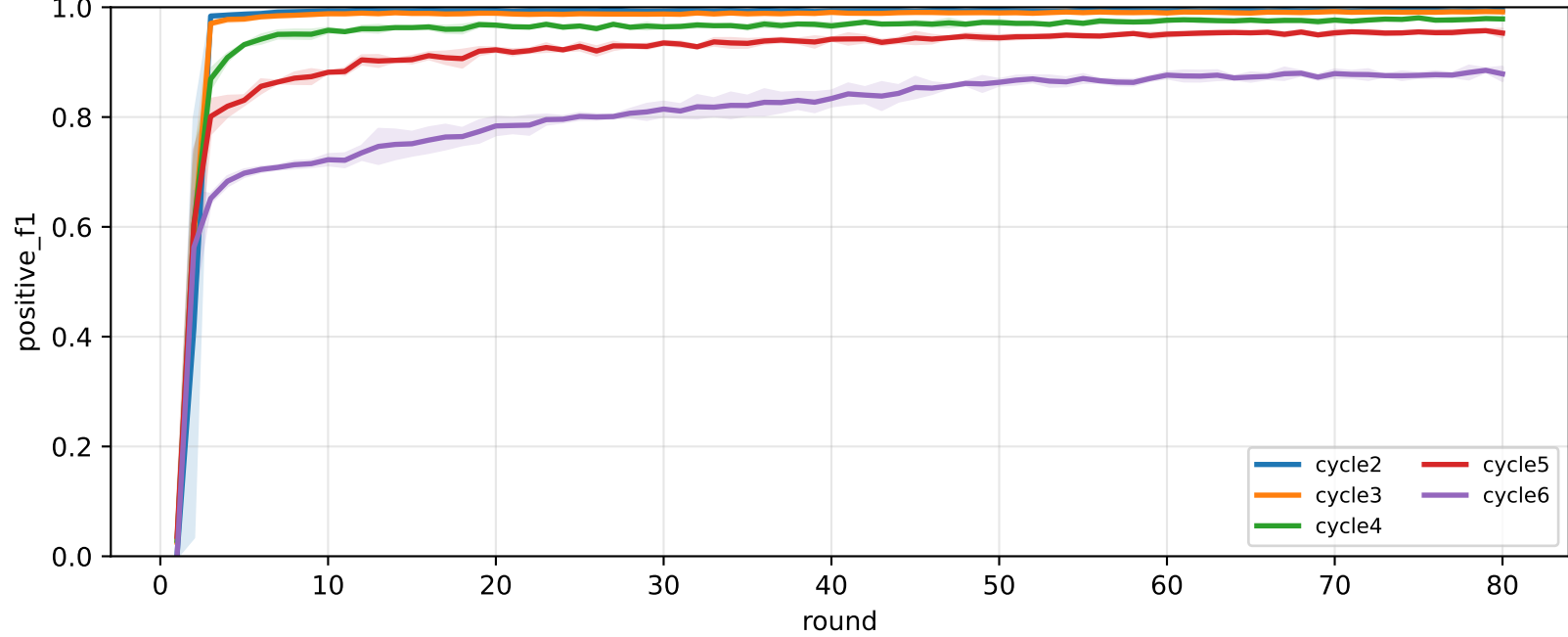
Validation loss comparison



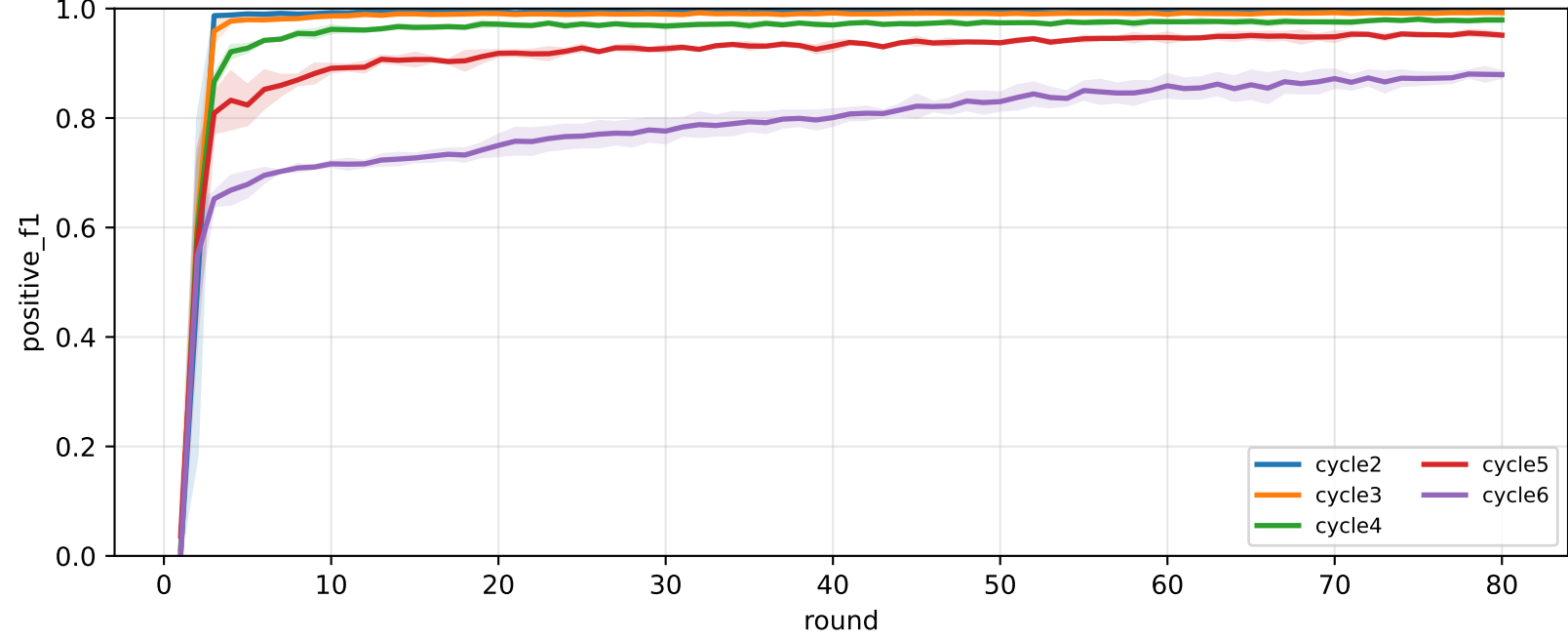
Pooled local centralized positive F1



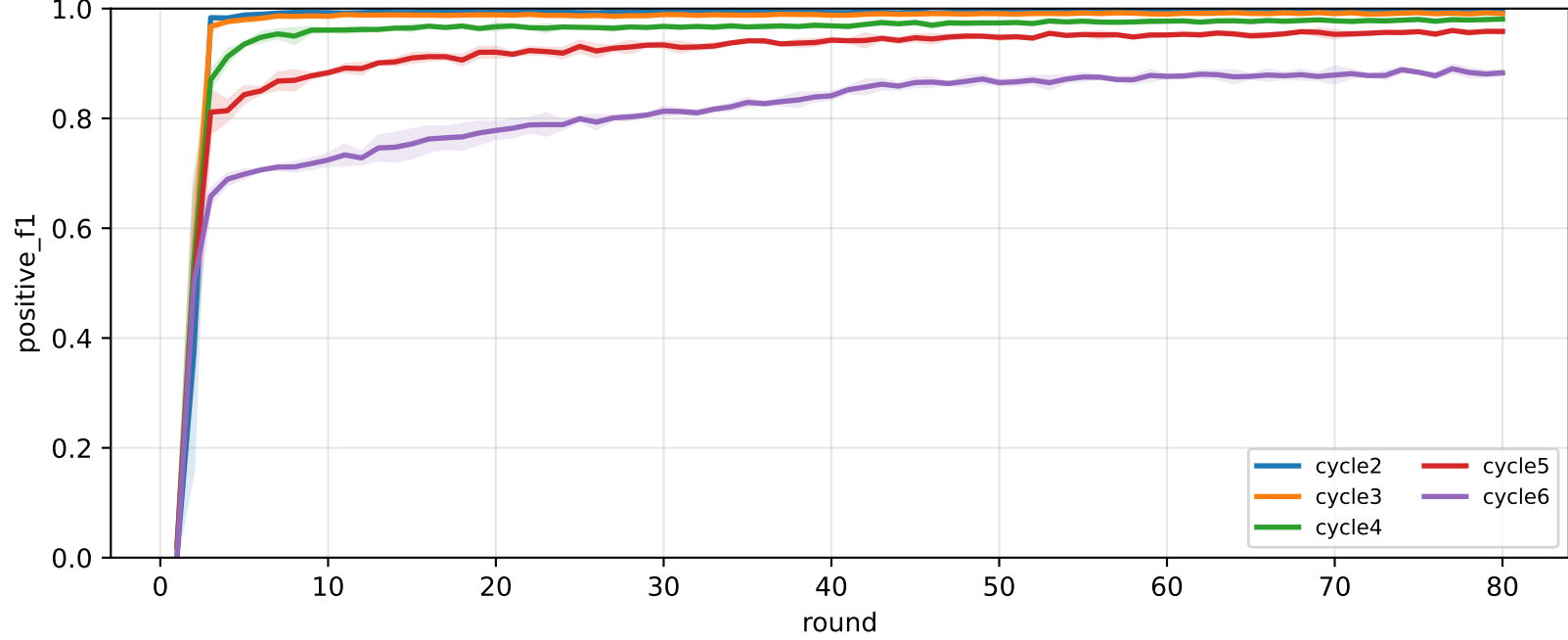
Pooled FedAvg positive F1



Pooled FedProx positive F1



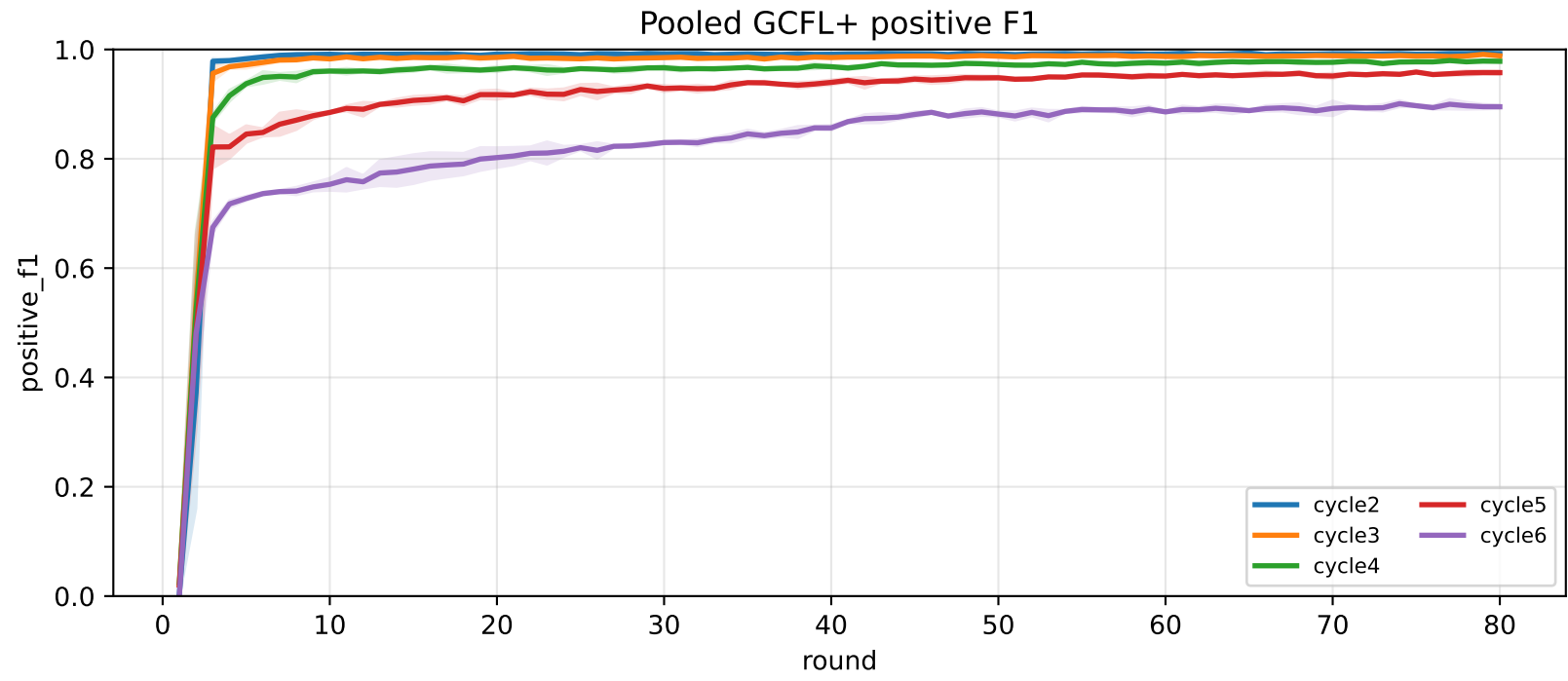
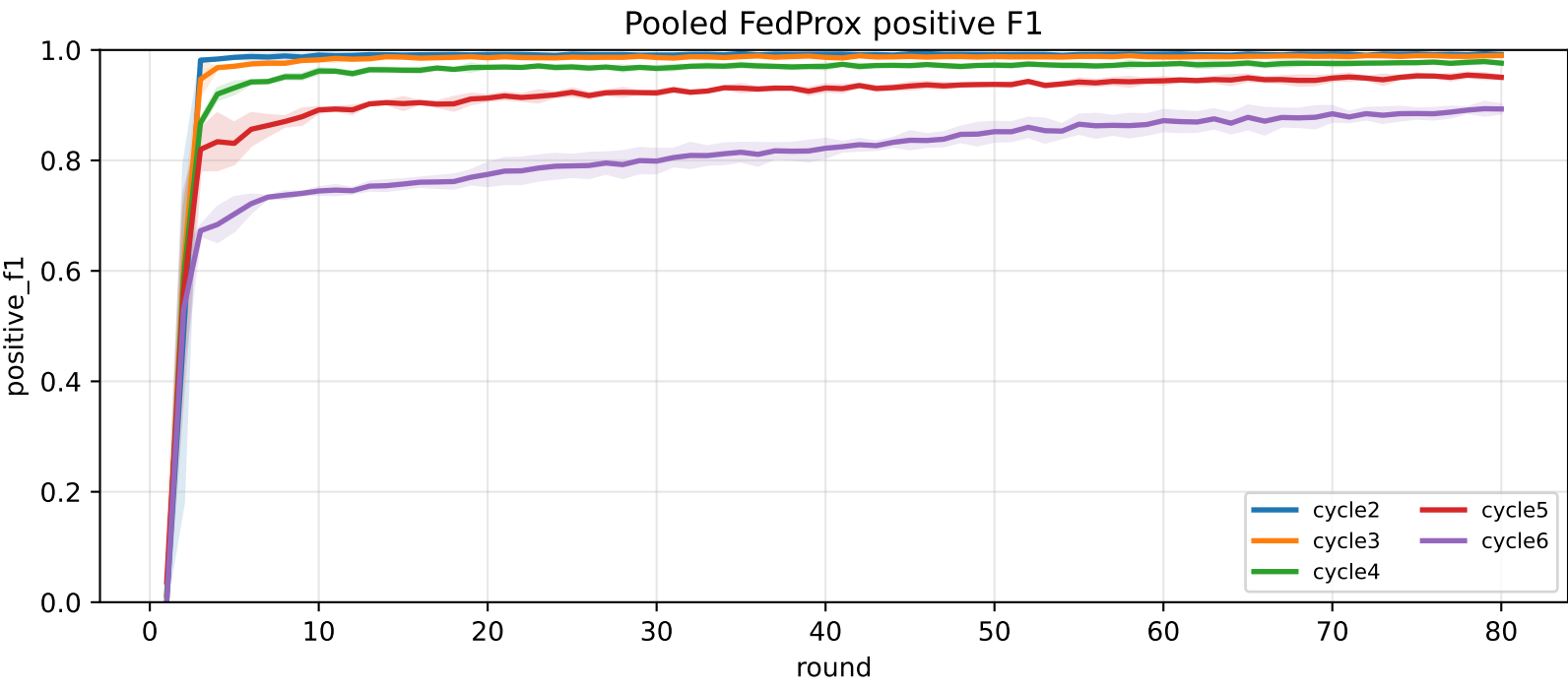
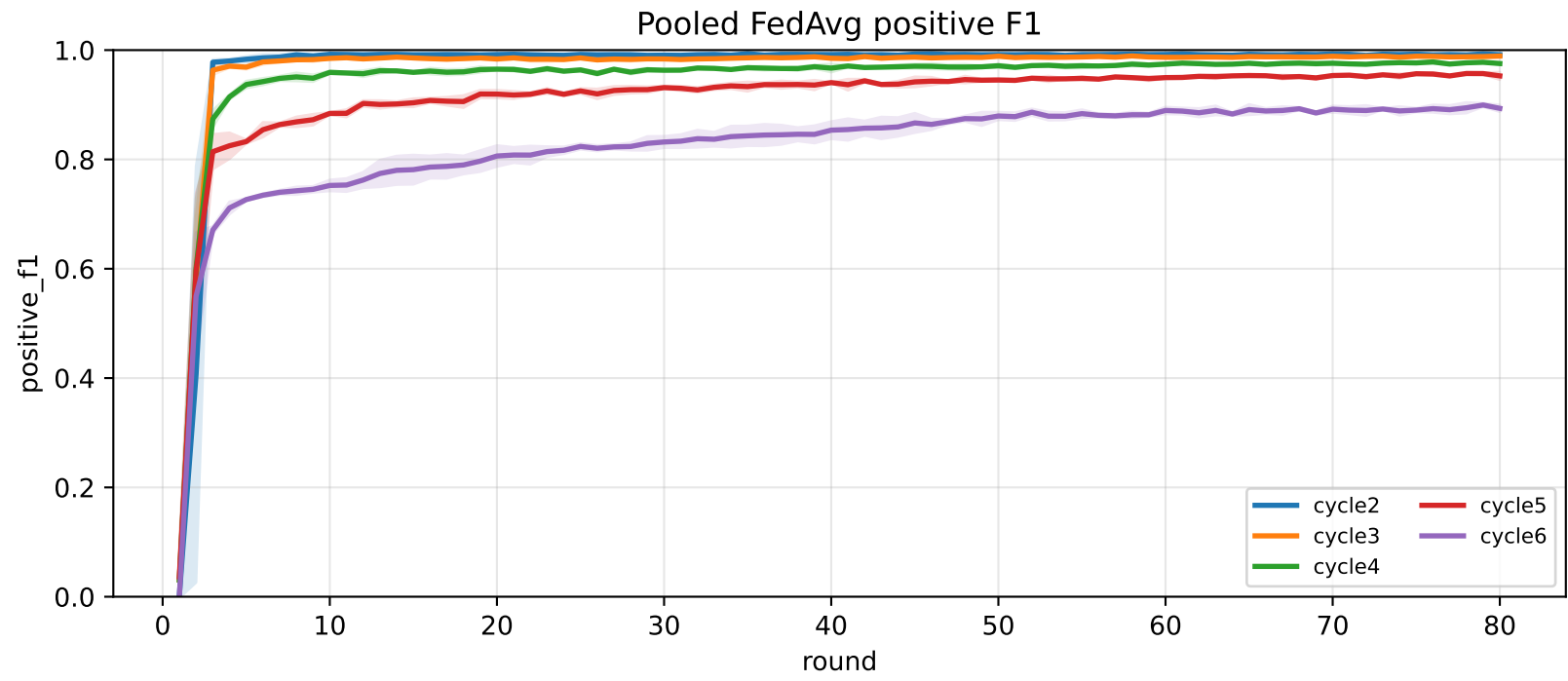
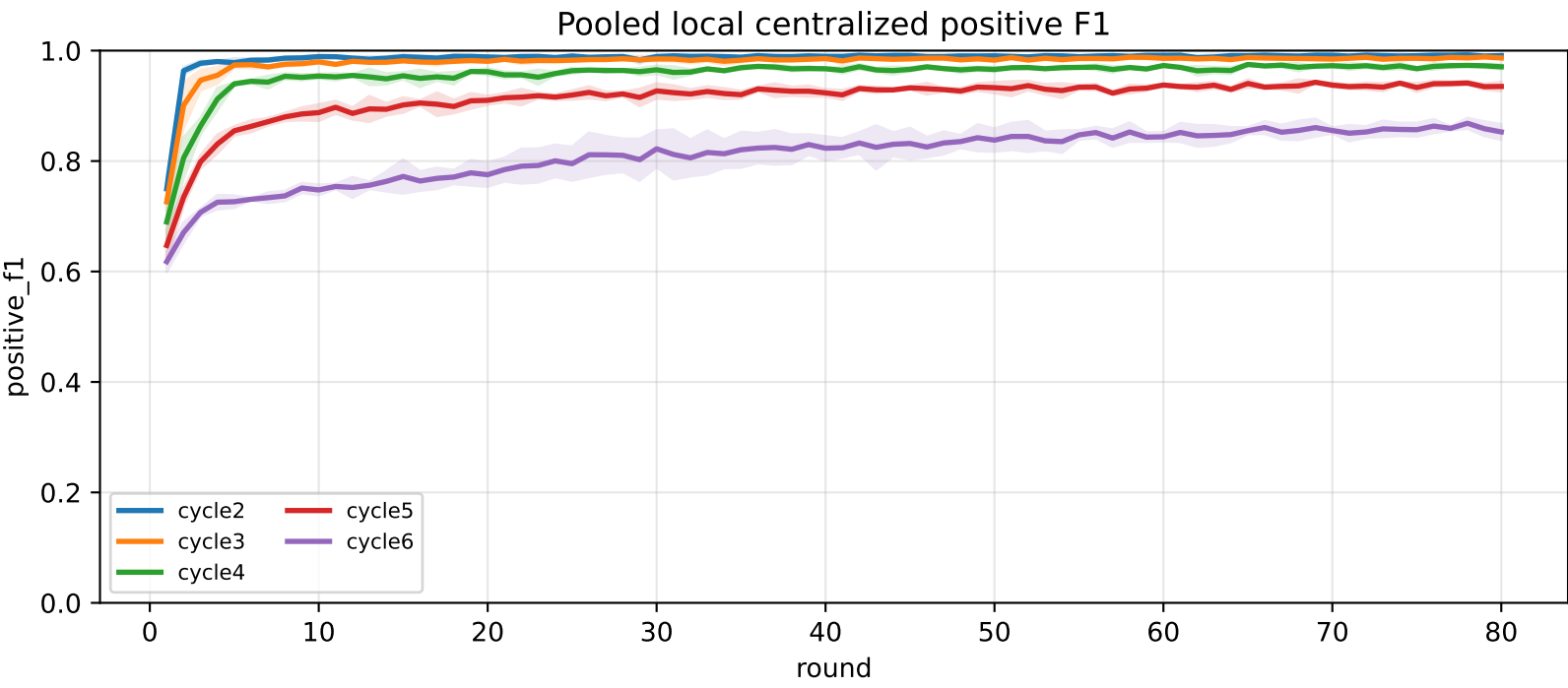
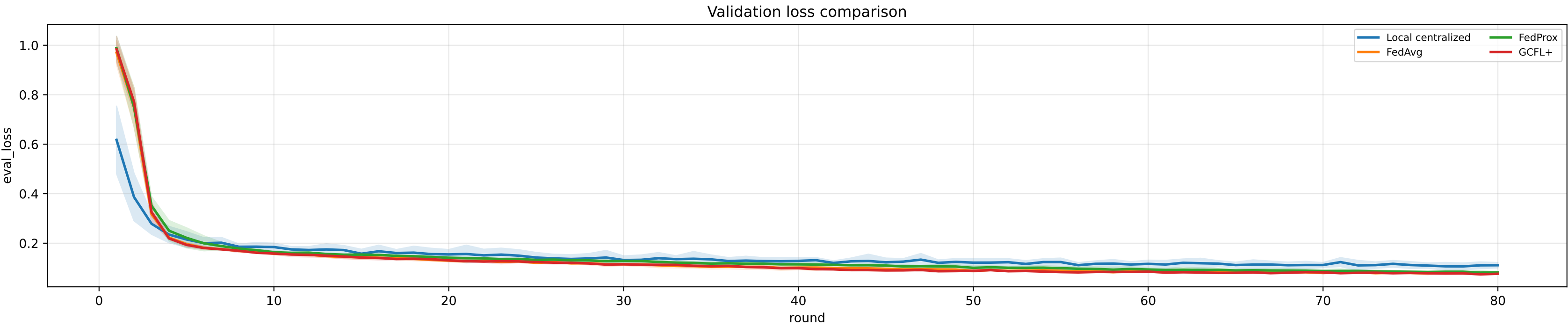
Pooled GCFL+ positive F1



Heterogeneity level: Low | family aggregated / pooled performance |
family=flat_balanced | n_clients=4 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=flat_balanced | cycle2=25.3%±0.6% | cycle3=26.4%±0.5% | cycle4=26.9%±0.4% | cycle5=26.6%±0.7% | cycle6=26.2%±0.9%
heterogeneity=low | task_jsd_mean=0.009 | task_jsd_median=0.004 | task_jsd_max=0.075 | family_centroid_jsd_mean=0.009
per-task pos-rate std | cycle2_std=0.073 | cycle3_std=0.042 | cycle4_std=0.021 | cycle5_std=0.032 | cycle6_std=0.050

Best logged test performance (mean ± std over seeds)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	95.0% ± 0.1%	95.2% ± 0.1%	99.2% ± 0.1%	98.8% ± 0.2%	97.4% ± 0.2%	94.1% ± 0.2%	86.6% ± 0.4%
FedAvg	96.3% ± 0.1%	96.4% ± 0.1%	99.2% ± 0.1%	98.7% ± 0.1%	97.7% ± 0.0%	95.9% ± 0.0%	90.5% ± 0.4%
FedProx	96.2% ± 0.3%	96.3% ± 0.2%	99.2% ± 0.1%	98.8% ± 0.1%	97.7% ± 0.2%	95.5% ± 0.4%	90.1% ± 0.7%
GCFL+	96.2% ± 0.1%	96.3% ± 0.1%	99.2% ± 0.0%	98.9% ± 0.1%	97.7% ± 0.1%	95.8% ± 0.2%	89.8% ± 0.4%



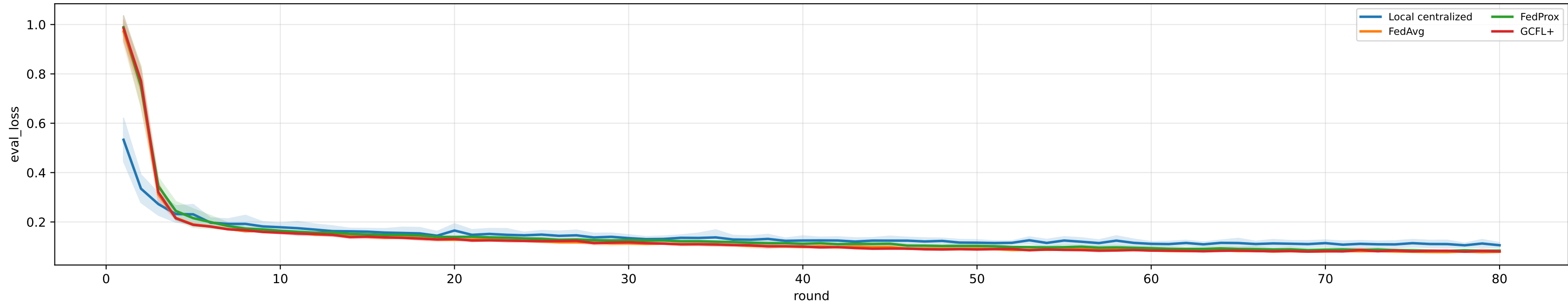
Heterogeneity level: Low | family aggregated / pooled performance |
family=mild_increasing | n_clients=4 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=mild_increasing | cycle2=22.3%±1.0% | cycle3=24.2%±0.4% | cycle4=25.3%±1.1% | cycle5=27.0%±0.5% | cycle6=27.3%±1.3%
heterogeneity=low | task_jsd_mean=0.009 | task_jsd_median=0.004 | task_jsd_max=0.075 | family_centroid_jsd_mean=0.009
per-task pos-rate std | cycle2_std=0.073 | cycle3_std=0.042 | cycle4_std=0.021 | cycle5_std=0.032 | cycle6_std=0.050

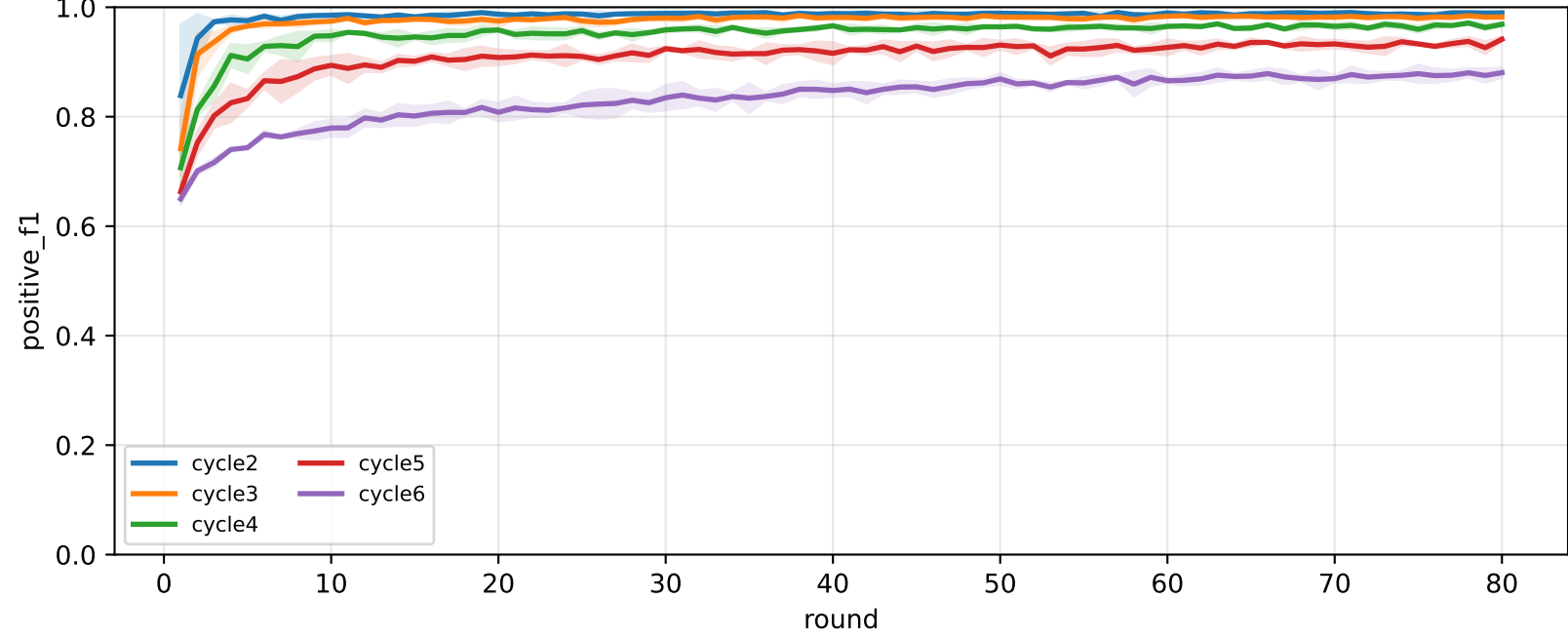
Best logged test performance (mean ± std over seeds)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	94.8% ± 0.3%	95.2% ± 0.3%	98.9% ± 0.1%	98.4% ± 0.2%	96.7% ± 0.3%	94.2% ± 0.2%	87.9% ± 0.9%
FedAvg	96.2% ± 0.1%	96.4% ± 0.1%	98.9% ± 0.2%	98.4% ± 0.1%	97.5% ± 0.1%	96.0% ± 0.1%	91.3% ± 0.1%
FedProx	96.0% ± 0.3%	96.2% ± 0.3%	99.0% ± 0.2%	98.5% ± 0.0%	97.4% ± 0.4%	95.5% ± 0.4%	90.9% ± 0.5%
GCFL+	95.7% ± 0.4%	96.0% ± 0.4%	99.0% ± 0.1%	98.4% ± 0.2%	97.2% ± 0.3%	95.4% ± 0.8%	90.1% ± 0.8%

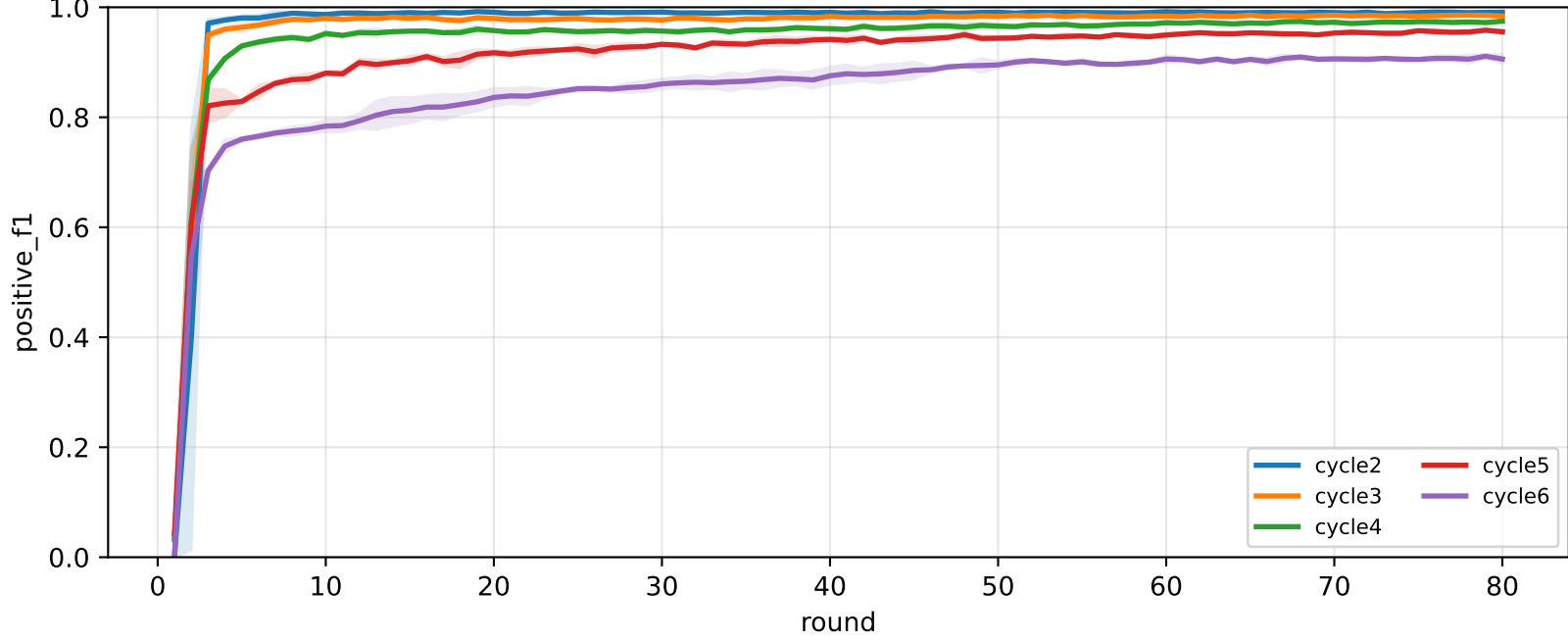
Validation loss comparison



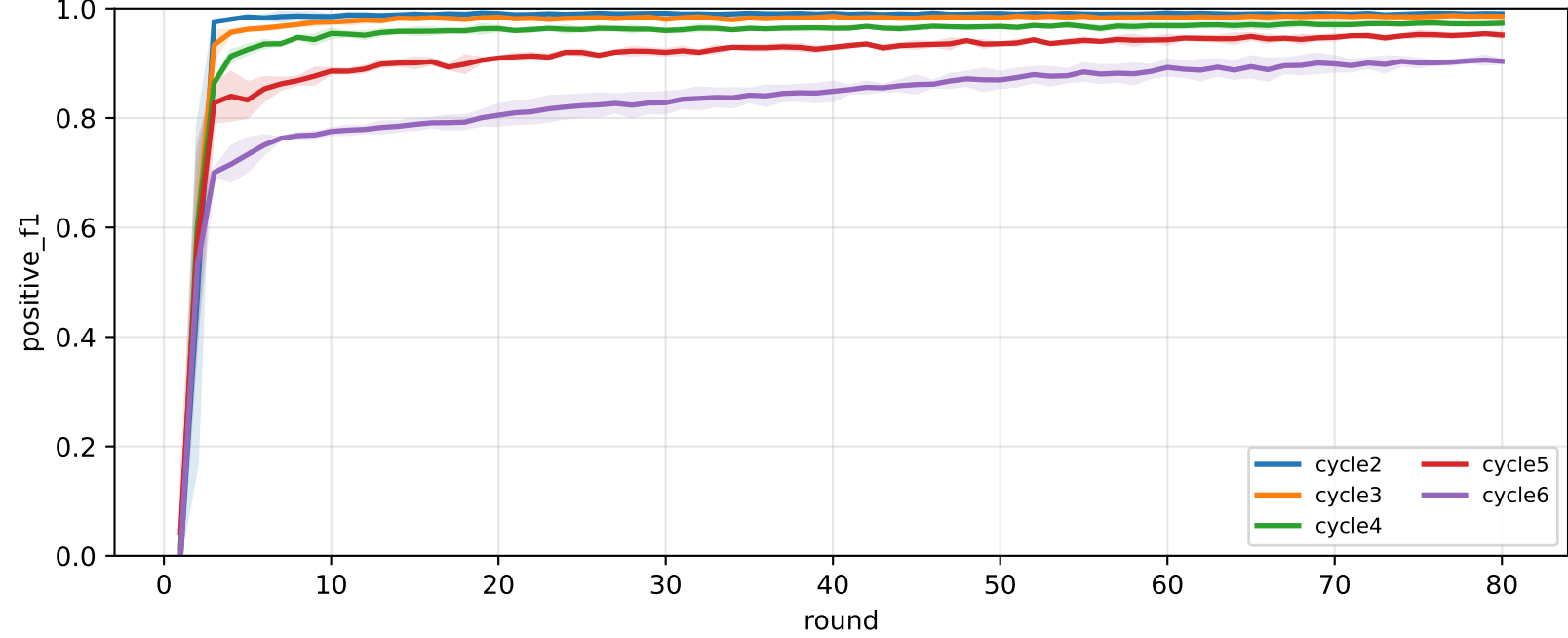
Pooled local centralized positive F1



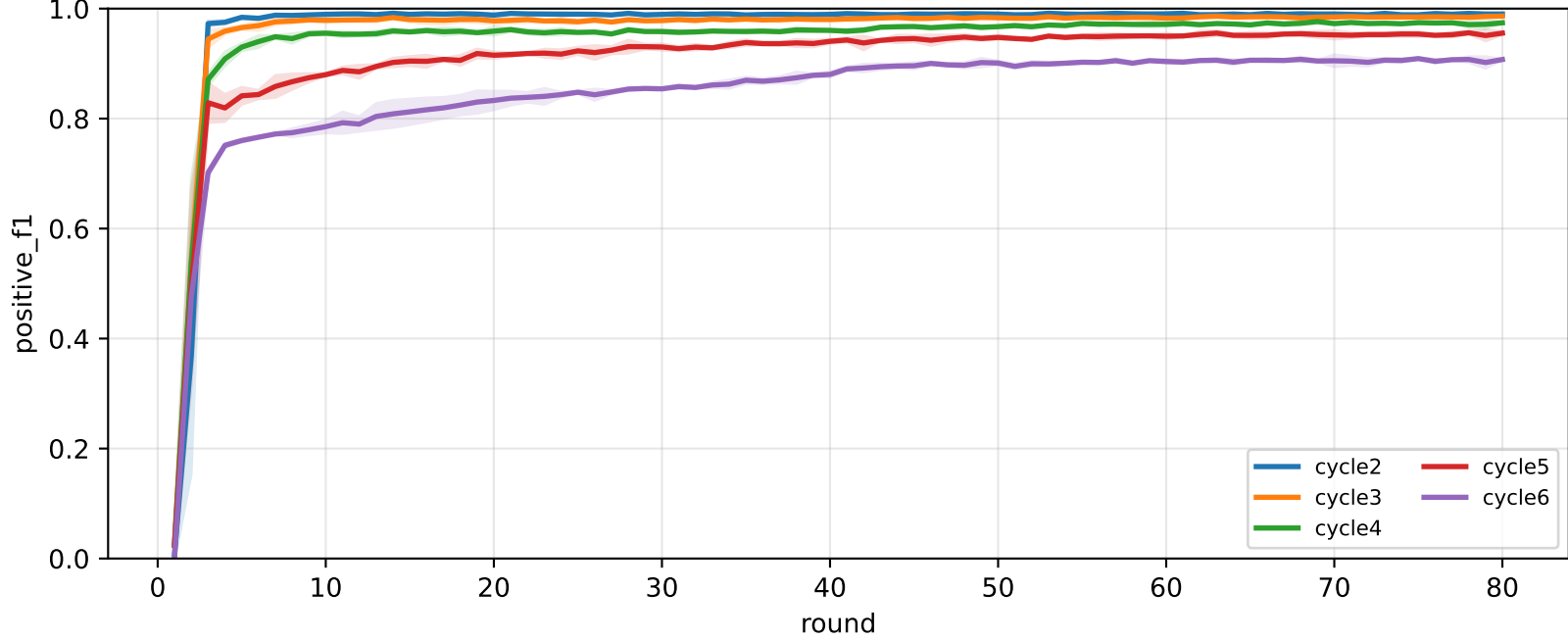
Pooled FedAvg positive F1



Pooled FedProx positive F1



Pooled GCFL+ positive F1



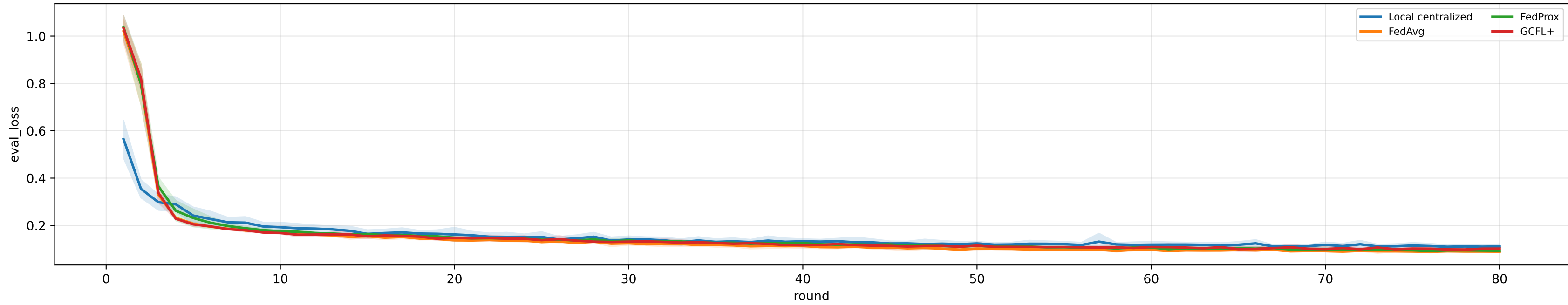
Heterogeneity level: Low | family aggregated / pooled performance |
family=strong_increasing | n_clients=4 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=strong_increasing | cycle2=15.0%±1.0% | cycle3=19.4%±0.7% | cycle4=22.9%±1.2% | cycle5=25.6%±1.2% | cycle6=27.6%±0.4%
heterogeneity=low | task_jsd_mean=0.009 | task_jsd_median=0.004 | task_jsd_max=0.075 | family_centroid_jsd_mean=0.009
per-task pos-rate std | cycle2_std=0.073 | cycle3_std=0.042 | cycle4_std=0.021 | cycle5_std=0.032 | cycle6_std=0.050

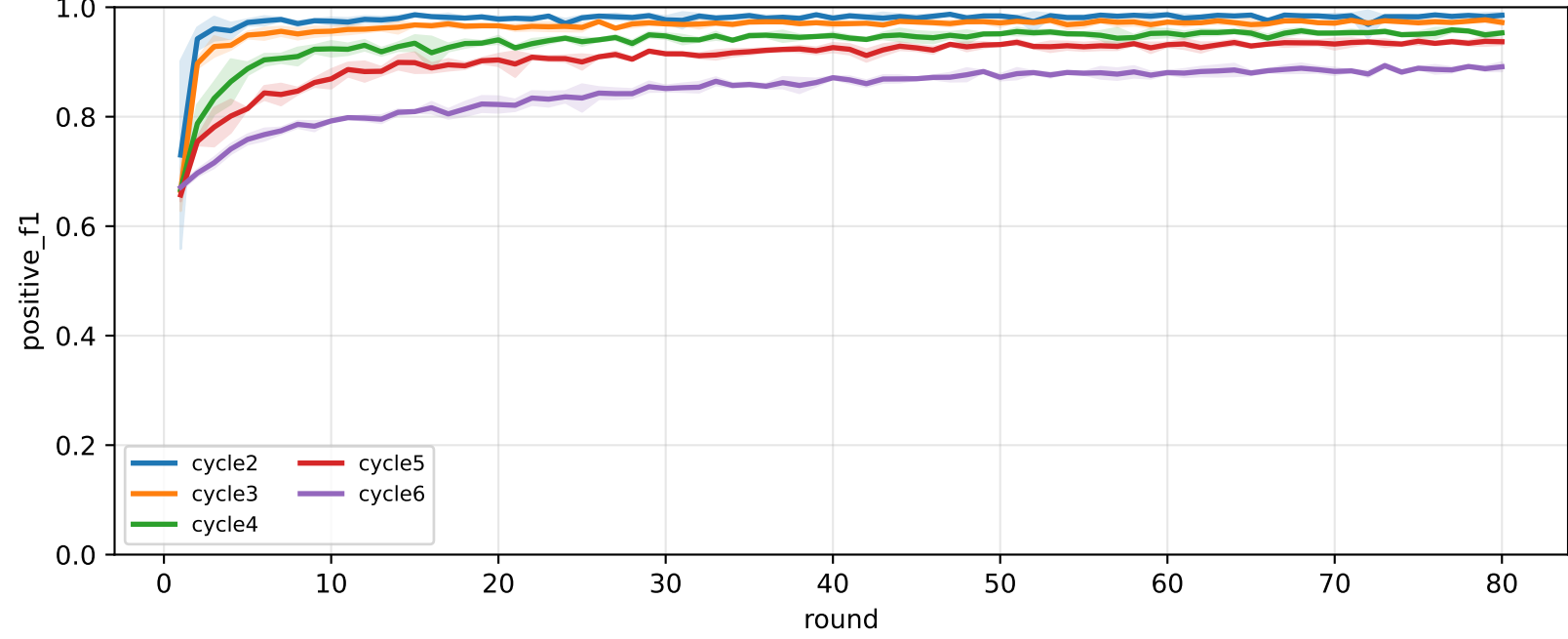
Best logged test performance (mean ± std over seeds)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	93.8% ± 0.2%	94.8% ± 0.2%	98.3% ± 0.3%	97.7% ± 0.2%	96.0% ± 0.1%	92.9% ± 0.3%	88.9% ± 0.5%
FedAvg	95.5% ± 0.1%	96.0% ± 0.1%	98.5% ± 0.2%	97.7% ± 0.3%	96.7% ± 0.1%	95.1% ± 0.4%	92.2% ± 0.1%
FedProx	95.2% ± 0.2%	95.8% ± 0.2%	98.5% ± 0.2%	97.8% ± 0.3%	96.3% ± 0.6%	94.5% ± 0.3%	92.0% ± 0.4%
GCFL+	94.7% ± 0.6%	95.4% ± 0.5%	98.6% ± 0.2%	97.5% ± 0.3%	95.9% ± 0.9%	94.3% ± 0.4%	91.0% ± 0.9%

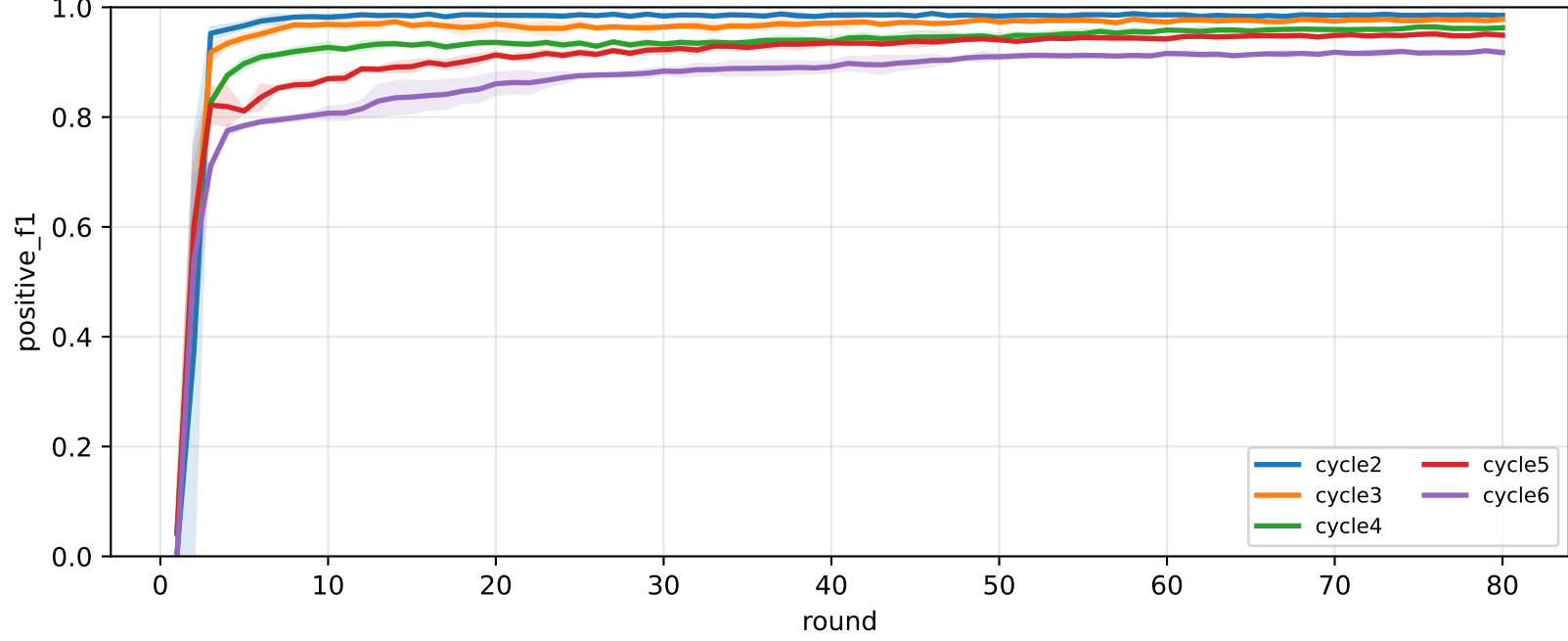
Validation loss comparison



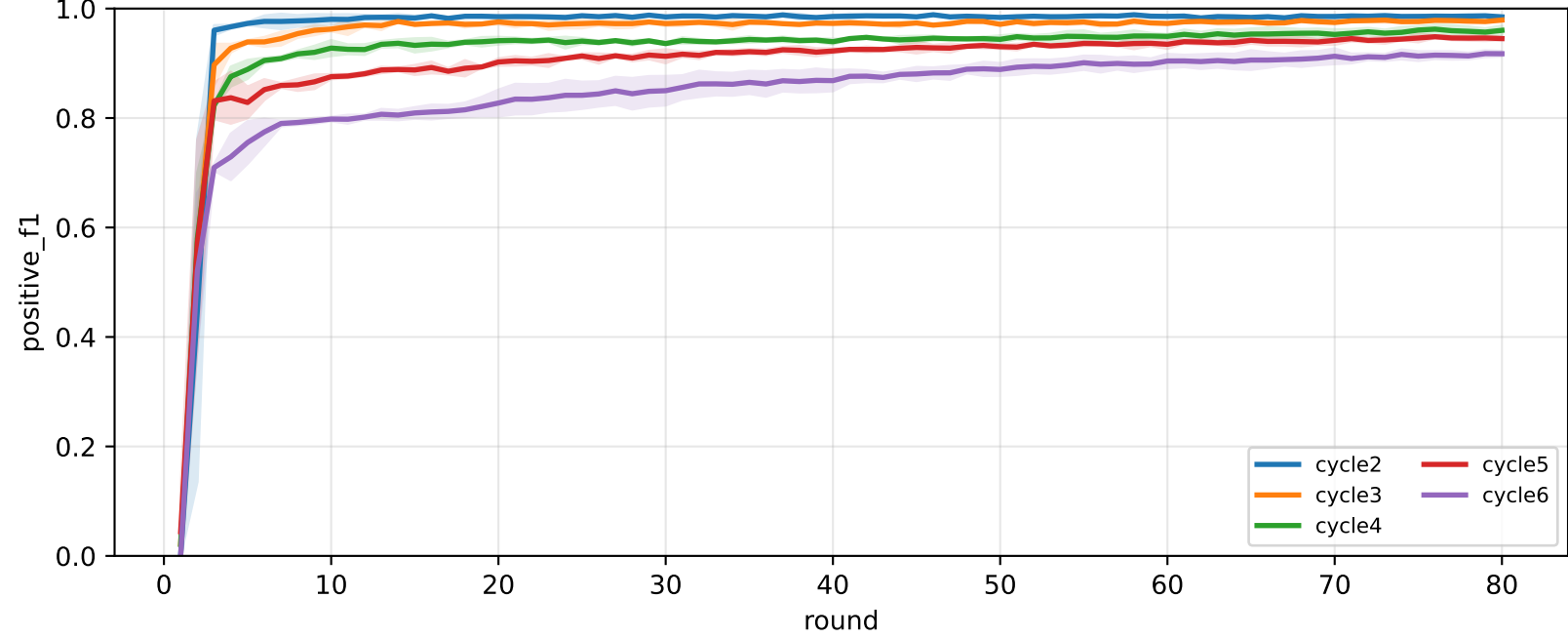
Pooled local centralized positive F1



Pooled FedAvg positive F1



Pooled FedProx positive F1



Pooled GCFL+ positive F1

